Prof. ZHANG, Eric Lu

張璐教授

BEng, MPhil, PhD
Associate Professor, Department of Computer Science

ericluzhang@comp.hkbu.edu.hk

DLB 629 +852 3411 5880

Personal Webpage HKBU Scholars

About

Dr. Lu (Eric) Zhang is an Associate Professor of Computer Science at Hong Kong Baptist University (HKBU). Before joining HKBU, he was a postdoctoral fellow in the Department of Computer Science and Pathology at Stanford University, supervised by Prof. Serafim Batzoglou and Prof. Arend Sidow. He received an MPhil degree from Li Ka Shing Faculty of Medicine at The University of Hong Kong and a Ph.D. degree in Computer Science from City University of Hong Kong in 2012 and 2016, respectively. In 2008, he received a B.Eng in Software Engineering from Tianjin University. In 2015, he was a visiting scholar in the Department of Mathematics at UC Berkeley and worked with Prof. Stephen Smale. He has an interdisciplinary background in genomics, statistics and computer science. His primary research interests are computational genomics, bioinformatics and AI for Genomics. His team works specifically on 1. Develop computational tools to support advanced high-throughput sequencing technologies on human gut metagenome. 2. Develop foundation models on metagenomics and single-cell multiomics data. 3. Explore the application of Large Language Models on modern healthcare problems. His work has been published in several top-tier journals, such as Nature Machine Intelligence, Nature Communications, Genome Biology, PNAS, Briefings in Bioinformatics, NAR Genomics and Bioinformatics, GigaScience, Nature Genetics, Bioinformatics, etc.

Research Interests

- Computational Genomics
- AI for Science
- Deep learning in Genomics
- Single-Cell Multiomics Sequencing
- Foundation model
- Large language model

Selected Publications

- Ke Xu, Yu Xu, Zirui Wang, Xin Maizie Zhou*, **Lu Zhang***. stDyer enables spatial domain clustering with dynamic graph embedding. **Genome Biology 2025.** (***Corresponding Author**)
- Bohao Zou, Jingjing Wang, Yi Ding, Zhenmiao Zhang, Yufen Huang, Xiaodong Fang, Ka Chun Cheung, Simon See, **Lu Zhang***. Deepurify: a multi-modal deep language model to remove contamination from metagenome-assembled genomes. **Nature Machine Intelligence 2024.** (***Corresponding Author**)
- Zhenmiao Zhang, Jin Xiao, Hongbo Wang, Chao Yang, Yufen Huang, Zhen Yue, Yang Chen, Lijuan Han, Kejing Yin, Aiping Lyu, Xiaodong Fang, **Lu Zhang***. Exploring high-quality microbial

genomes by assembling short-reads with long-range connectivity. **Nature Communications 2024.** (***Corresponding Author**)

- Yu Xu, Jiaxing Chen, Aiping Lyu*****, William K Cheung*****, **Lu Zhang***. dynDeepDRIM: a dynamic deep learning model to infer direct regulatory interactions using single cell time-course gene expression data. **Briefings in Bioinformatics** 2022**.** (***Corresponding Author**)
- Jiaxing Chen, Chinwang Cheong, Liang Lan, Xin Zhou, Jiming Liu, Aiping Lyu, William K Cheung*, **Lu Zhang***. DeepDRIM: a deep neural network to reconstruct cell-type-specific gene regulatory network using single-cell RNA-Seq Data. **RECOMB-Seq 2021** and **Briefings in Bioinformatics** 2021. (***Corresponding Author**)
- Xin Zhou, **Lu Zhang**, Ziming Weng, David L. Dill, Arend Sidow. Aquila enables reference-assisted diploid personal genome assembly and comprehensive variant detection based on linked reads. **Nature Communications** 2021.
- **Lu Zhang***, Xiaodong Fang et al. A Comprehensive Investigation of Metagenome Assembly by Linked-Read Sequencing. **Microbiome**. (***Corresponding Author**)
- **Lu Zhang**, Xin Zhou, Ziming Weng, Arend Sidow. De novo diploid genome assembly for genome-wide structural variant detection. **NAR Genomics and Bioinformatics** 2020.
- **Lu Zhang**, Xin Zhou, Ziming Weng, Arend Sidow. Assessment of human diploid genome assembly with 10x Linked-Reads data. **GigaScience** 2019.
- JiFeng Guo†, **Lu Zhang†** et al. De novo coding mutations contribute to early onset Parkinson's disease. **Proceedings of the National Academy of Sciences of the United States of America**, 2018 (**Joint First Author**).
- Xin Zhou, Serafim Batzoglou, Arend Sidow, **Lu Zhang***. HAPDeNovo: a haplotype-based approach for filtering and phasing de novo mutations in linked read sequencing data. **BMC Genomics,** 2018 (**Corresponding Author**).
- Xueyan Li†, Dingding Fan†, Wei Zhang†, Guichun Liu†, **Lu Zhang†**, et al. Outbred genome sequencing and CRISPR/Cas9 gene editing in butterflies. **Nature Communications**, 2015 (**Joint First Author**).